



# California Regional Water Quality Control Board

## Central Coast Region



Linda S. Adams  
Secretary for  
Environmental  
Protection

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Arnold Schwarzenegger  
Governor

July 31, 2008

Clifton W. Price  
Public Works Director  
City of Soledad  
P.O. Box 156  
Soledad, CA 93960

Dear Mr. Price:

### **ADOPTED MASTER WATER RECLAMATION REQUIREMENTS FOR THE CITY OF SOLEDAD, MONTEREY COUNTY**

Enclosed are **Master Water Reclamation Requirements** (Order No. R3-2008-0042) for the City of Soledad, which are effective immediately. The Central Coast Water Board adopted the **Master Water Reclamation Requirements** on May, 9, 2008.

If you have any questions or comments, please contact **Tom Kukol at (805) 549-3689,** email **tkukol@waterboards.ca.gov,** or call Burton Chadwick at (805) 542-4786.

Sincerely,

Roger W. Briggs  
Executive Officer

Enclosure: Order No. R3-2008-0042

cc: City of Soledad IPL

Task: 12601  
File: City of Soledad Wastewater Treatment Facilities\WDR\WDR Facilities\Monterey Co\City of Soledad WWTP\Reclamation Requirements\draft transmittal.doc





**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL COAST REGION  
895 Aerovista Place, Suite 101  
San Luis Obispo, CA 93401-7906**

**ORDER NO. R3-2008-0042**

**MASTER RECLAMATION PERMIT  
FOR  
THE CITY OF SOLEDAD  
WATER RECYCLING FACILITY  
MONTEREY COUNTY**

**(Waste Discharger Identification No. 3 270113001)**

The California Regional Water Quality Control Board, Central Coast Region (hereafter "Water Board"), finds:

**IMPORTANCE OF RECYCLED WATER**

1. California Water Code Section 13510 states that the people of the state have a primary interest in the development of facilities to recycle water containing waste to supplement existing surface and underground water supplies and to assist in meeting the future water requirements of the state.
2. California Water Code Section 13512 states that it is the intention of the legislature that the State undertake all possible steps to encourage development of water recycling facilities so that recycled water may be made available to help meet the growing water demands of the State.

**FACILITY INFORMATION**

3. The City of Soledad (hereafter "Discharger," "Supplier," "Distributor," or "Supplier and Distributor") provides wastewater collection, treatment, and disposal services for City customers, including two California Department of Corrections prison facilities.
4. Wastewater treatment and disposal occurs at two distinct locations; the "City Plant" and the "Prison Plant," as described by Attachment "A."
  - a. The City Owns and operates the City Plant.
  - b. The City Leases and operates Prison Plant, but is pursuing ownership.
5. On December 21, 2007, the City submitted a *Report of Waste Discharge* as an application for a Master Reclamation Permit. The *Report of Waste Discharge* proposes wastewater service improvements that will provide for water recycling. Specifically, the City proposes the following:
  - a. Upgrade the City Plant from a 3.0 MGD oxidation pond system to a 5.5 MGD activated sludge system with tertiary filtration and disinfection.
  - b. Connect the City Plant to the Prison Plant via a 5-mile transmission/distribution pipeline.
  - c. Construct a 2.6 MGD "Scalping Plant" near the City's *Miravale III* expansion area to provide recycled water to that area and to provide additional disposal.

- d. Maintain existing disposal capacities at the City Plant (4.3 MGD) and the Prison Plant (1.2 MGD).

#### **INTENT OF THIS ORDER**

6. California Water Code Section 13523.1 provides that a Water Board may issue a master reclamation permit to a supplier or distributor, or both, of reclaimed water.
7. This Order is intended to serve as a master reclamation permit that is consistent with California Water Code Section 13523.1.
8. This Order is intended to supersede Order No. R3-2005-0074, adopted by the Water Board on May 13, 2005.

#### **BASIN PLAN**

9. The Water Quality Control Plan, Central Coast Basin (Basin Plan) was adopted by the Water Board on November 19, 1989, and approved by the State Water Resources Control Board (State Water Board) on August 16, 1990. The Water Board approved amendments to the Basin Plan on February 11, 1994, and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State Waters. This Order implements the Basin Plan.
10. The Basin Plan designates the existing and anticipated beneficial uses of groundwater in the vicinity of the land disposal discharge to include:
  - a. Municipal and Domestic Water Supply;
  - b. Agricultural Water Supply
  - c. Industrial Process Supply; and,
  - d. Industrial Service Supply.
11. The Salinas River is the closest surface water body to the Facility and reuse areas. The Basin Plan designates existing and anticipated beneficial uses of the Salinas River along the reach adjacent to the Facility and reuse areas (Chualar to Nacimiento River) that could be affected by the discharge to include:
  - a. Municipal and Domestic Supply;
  - b. Agricultural Water Supply;
  - c. Industrial Process Supply;
  - d. Industrial Service Supply;
  - e. Groundwater Recharge;
  - f. Water Contact Recreation;
  - g. Non-Contact Water Recreation;
  - h. Wildlife Habitat;
  - i. Cold Freshwater Habitat;
  - j. Warm Freshwater Habitat;
  - k. Migration of Aquatic Organisms;
  - l. Spawning, Reproduction, and/or Early Development
  - m. Rare, Threatened, or Endangered Species
  - n. Commercial and Sport Fishing.
12. This master reclamation permit implements the Basin Plan's water quality objectives.

**ANTI-DEGRADATION**

13. Antidegradation: State Water Board Resolution No. 68-16 *Statement of Policy with Respect to Maintaining High Quality of Waters in California* (Resolution No. 68-16) requires Regional Water Boards, in regulating the discharge of waste, to maintain high quality waters of the State until it is demonstrated that any change in quality will be consistent with maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in a Regional Water Board's policies (e.g., quality that exceeds applicable water quality standards). Resolution No. 68-16 also states, in part:

*Any activity which produces or may produce a waste or increased volume or concentration of waste and which discharges or proposes to discharge to existing high quality waters will be required to meet waste discharge requirements which will result in best practicable treatment and control of the discharge necessary to assure that (a) a pollution or nuisance will not occur and (b) the highest water quality consistent with maximum benefit to the people of the State will be maintained.*

The discharges regulated by this Order are subject to waste discharge requirements that will result in best practicable treatment or control, the prevention of pollution and nuisance, and maintenance of the highest water quality consistent with maximum benefit to the people of the State.

**TMDL**

14. The Salinas River and several of its tributaries are on the Clean Water Act Section 303(d) list as impaired due to elevated concentrations of nutrients and pathogens. Waste load and load allocations will be developed for sources of nutrients and pathogens in the Salinas River, as well as other water bodies within the Salinas River watershed. If the Water Board determines that discharges from the Facility are causing or contributing to nutrient or fecal coliform related water quality impairment, waste discharges described in this Order may be modified to meet the allocations described in proposed future TMDLs.

**STORMWATER**

15. Stormwater runoff from the Discharger's treatment, disposal, and reuse areas is regulated under separate order.

**COLLECTION SYSTEM**

16. The Discharger's collection system is regulated under separate order, the *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems*, Order No. 2006-0003 (General Order).

**ENVIRONMENTAL REVIEW**

17. On September 21, 2005, the City of Soledad certified a final Environmental Impact Report in accordance with the California Environmental Quality Act (Public Resources Code, Section 621000 et seq.) and the California Code of Regulations. The City of Soledad has determined there are no significant adverse environmental effects or that all potentially significant adverse effects can be avoided through implementation of mitigation measures. Mitigation measures to prevent nuisance and ensure protection of beneficial uses of surface water and groundwater will be implemented through this order.

**GENERAL FINDINGS**

18. No discharge of waste to waters of the state creates a vested right to continue the discharge. All discharges of waste into waters of the state are privileges, not rights. A permit is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and of the Clean Water Act (as amended or as supplemented by implementing guidelines and regulations) and requirements necessary to implement water quality control plans, protect beneficial uses, and prevent nuisance. Compliance with this Order should ensure that water quality is protected.
19. On April 17, 2008, the Water Board notified the Discharger and other interested parties of its intent to prescribe Supplier and Distributor master water reclamation requirements for the Facility and associated reuse areas, respectively. In addition, the Water Board provided the public with an opportunity for a public hearing and the opportunity to submit written comments.
20. The requirements of this Order conform with and implement the water reclamation criteria of the State Department of Public Health (Title 22) to protect the public health, safety, and welfare. The Water Board has consulted with, and received the recommendations from, the Department of Public Health regarding the regulation of this discharge.
21. The Water Board, at a public meeting held July 11, 2008, heard and considered all comments pertaining to the discharge and found this Order consistent with the above findings.
22. Any person affected by this action of the Water Board may petition the State Water Board to review the action in accordance with Section 13320 of the California Water Code and Title 23 of the California Code of Regulations, Section 2050 et seq. The State Water Board must receive the petition within 30 days of the date of this Order. Copies of the law and regulations applicable to filing petitions will be provided upon request.
23. Requirements specified in this Order are intended to ensure proper treatment and handling of recycled domestic wastewater for the protection of public health. In addition, recycled water treatment and storage requirements of this Order in conjunction with the requirements for the application of the recycled wastewater in designated reuse areas does not pose a significant threat to surface water or underlying groundwater resources.

**IT IS HEREBY ORDERED**, pursuant to authority in Sections 13263 and 13523.1 of the California Water Code, that the City of Soledad, its agents, successors, and assigns, may produce, store and distribute reclaimed wastewater provided it complies with the following:<sup>1</sup>

Throughout these requirements footnotes are listed to indicate the source of requirements specified. Requirement footnotes are as follows (requirements without footnotes are BPJ unless otherwise noted). Numbered footnotes generally reference code sections for direct citations:

BPJ

Best Professional Judgment of Regional Water Quality Control Board Staff

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<sup>1</sup> General permit conditions, definitions and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January 1984, included as part of this Order.

|       |                                                                  |
|-------|------------------------------------------------------------------|
| ROWD  | The Discharger's Report of Waste Discharge                       |
| 40CFR | Title 40 Code of Federal Regulations                             |
| BP    | Central Coast Regional Water Quality Control Plan                |
| DPH   | California Department of Public Health                           |
| T22   | Title 22 CCR, Division 4, Chapter 3, Water Reclamation Criteria  |
| CWC   | Porter-Cologne Water Quality Control Act (California Water Code) |

The Supplier and Distributor shall comply with all Prohibitions, Specification, and Provisions as applicable, and shall ensure that indirect Users also comply with these requirements. The Supplier and Distributor shall comply with the specific Supplier Requirements and Distributor/User Requirements, respectively. The Distributor shall ensure that indirect Users also comply with applicable Distributor/User Requirements.

## A. PROHIBITIONS

1. Discharge of treated wastewater to areas other than disposal areas, or areas of authorized storage and use, is prohibited.<sup>ROWD, BPJ</sup>
2. Discharge of untreated or partially treated wastes; including overflows, bypasses, seepages, and spills; is prohibited.<sup>BPJ, PC</sup>
3. The treatment, storage, distribution, or reuse of recycled water shall not create a nuisance as defined in section 13050(m) of the California Water Code.<sup>CWC</sup>
4. No recycled water used for irrigation shall be applied during periods of rainfall or when soils are saturated such that runoff occurs.<sup>BPJ</sup>
5. No recycled water shall be discharged from the treatment facilities, storage ponds, or other containment, other than for designated irrigation or other approved reuse applications in accordance with this Order.<sup>BPJ</sup>
6. The incidental discharge of recycled water to waters of the State shall not unreasonably affect the beneficial uses of the water, and not result in an exceedance of an applicable water quality objective in the receiving water.<sup>2 BPJ</sup>
7. There shall be no cross-connections between the potable water supply and pipes containing recycled water. Supplementing recycled water with water used for domestic supply shall not be allowed except through an air-gap separation. In accordance with California Code of Regulations (CCR) Title 17, Section 7604, a reduced pressure principle backflow device shall be provided at premises where recycled water is used and there is no interconnection with the potable water system.<sup>3, BPJ, T22</sup>

## B. SPECIFICATIONS

### Flow and General Limitations

<sup>2</sup> February 24, 2004, SWRCB memorandum re: Incidental Runoff of Recycled Water.

<sup>3</sup> This requirement does not apply to individual residences using recycled water for landscape irrigation as part of an approved dual plumbed use area as defined in CCR Title 17, Table 1 Section (c)(3).

1. Monthly average influent wastewater flow to the City Plant shall not exceed 5.5 MGD. <sup>ROWD, BPJ</sup>
2. Monthly average influent wastewater flow to the Scalping Plant shall not exceed 2.6 MGD. <sup>ROWD, BPJ</sup>
3. Monthly average treated wastewater flow to the City Plant percolation basins shall not exceed 4.3 MGD. <sup>ROWD, BPJ</sup>
4. Monthly average treated wastewater flow to the Prison Plant percolation basins shall not exceed 1.2 MGD. <sup>ROWD, BPJ</sup>
5. Through January 2010, the Facility effluent shall not exceed the following effluent limitations:

**Effluent Limitations (mg/L)**

| Parameter                           | Daily Max | 30-Day Mean |
|-------------------------------------|-----------|-------------|
| BOD <sub>5</sub> <sup>a</sup>       | 30        | --          |
| Total Suspended Solids <sup>a</sup> | 30        | --          |
| Ammonia, as N <sup>a</sup>          | --        | --          |
| Nitrate as N <sup>a</sup>           | --        | --          |
| Total Dissolved Solids <sup>b</sup> | --        | 990         |
| Sodium <sup>b</sup>                 | --        | 175         |
| Chloride <sup>b</sup>               | --        | 250         |
| Boron <sup>b</sup>                  | --        | 0.5         |
| Sulfate <sup>b</sup>                | --        | 205         |

Notes:

a. As measured after disinfection and prior to the effluent storage basins

b. As delivered concentrations measured in the second effluent storage pond

6. After January 2010, the Facility effluent shall not exceed the following effluent limitations:

**Effluent Limitations (mg/L)**

| Parameter                           | Daily Max | 30-Day Mean |
|-------------------------------------|-----------|-------------|
| BOD <sub>5</sub> <sup>a</sup>       | 10        | --          |
| Total Suspended Solids <sup>a</sup> | 10        | --          |
| Ammonia, as N <sup>a</sup>          | 5         | --          |
| Nitrate as N <sup>a</sup>           | 5         | --          |
| Total Dissolved Solids <sup>b</sup> | --        | 990         |
| Sodium <sup>b</sup>                 | --        | 175         |
| Chloride <sup>b</sup>               | --        | 250         |
| Boron <sup>b</sup>                  | --        | 0.5         |
| Sulfate <sup>b</sup>                | --        | 205         |

Notes:

c. As measured after disinfection and prior to the effluent storage basins

d. As delivered concentrations measured in the second effluent storage pond

7. The effluent pH shall not be less than 6.5 or greater than 8.4. <sup>BP</sup>

**Disinfected Tertiary Recycled Water Limitations**

8. The Supplier shall ensure that treated effluent put to use for disinfected tertiary recycled water applications shall be an adequately oxidized, filtered, and disinfected water, as defined in CCR Title 22, Division 4, Chapter 3, Sections 60301-60335, or alternatively defined and approved by State Department of Public Health (DPH).
9. The turbidity of the disinfected tertiary recycled water shall not exceed any of the following:<sup>4, 5, 6</sup>
  - a. An average of 2 nephelometric turbidity units (NTU) within a 24-hour period;
  - b. 5 NTU more than 5 percent of the time within a 24-hour period; and
  - c. 10 NTU at any time.
10. Disinfected tertiary recycled water shall not contain total coliform concentrations exceeding the following limits:<sup>7</sup>
  - a. the seven-day median concentration must not exceed an MPN of 2.2/100 ml; and
  - b. concentrations must not exceed 23/100 ml in more than one sample taken over a 30-day range;
  - c. concentrations must be less than 240/100 ml at all times.
11. For chlorine-based disinfection, the chlorine residual within the disinfection process following filtration shall provide a CT value<sup>8</sup> of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow. Ultraviolet disinfection must maintain compliance with the Water Research Institute's (NWRI) "UV Disinfection Guidelines for Wastewater Reclamation in California."<sup>9, 10</sup>

**C. SUPPLIER AND DISTRIBUTOR REQUIREMENTS**

1. Reclamation facilities shall be operated in conformance with the California Department of Public Health "Guidelines for Use of Reclaimed Wastewater for Irrigation and Impoundment," "Guidelines for Worker Protection at Reclamation Use Areas," the American Water Works Association, California-Nevada Section's *Guidelines for the Distribution of Non-potable Water*, and the Distributor's approved reclaimed water use rules and regulations (which may clarify and/or modify the above guidelines) and the appropriate local administrative procedures.

<sup>4</sup> CCR Title 22, Div. 4, Chap.3, Section 60301.320

<sup>5</sup> Pursuant to CCR Title 22, Div. 4, Chap.3, Section 60301.320(a) coagulation need not be used as part of the treatment process provided that the filter effluent turbidity does not exceed 2 NTU, the turbidity of the influent to the filters is continuously measured, the influent turbidity does not exceed 5 NTU for more than 15 minutes and never exceeds 10 NTU, and that there is the capability to automatically activate chemical addition or divert the wastewater should the filter influent turbidity exceed 5 NTU for more than 15 minutes.

<sup>6</sup> Pursuant to CCR Title 22, Div. 4, Chap.3, Section 60301.320(b) for filtration via microfiltration, ultrafiltration, nonofiltration, or reverse osmosis the effluent turbidity shall not exceed 0.2 NTU more than 5 percent of the time within a 24-hour period, and 0.5 NTU at any time.

<sup>7</sup> CCR Title 22, Div. 4, Chap.3, Section 60301.230

<sup>8</sup> The product of total chlorine residual and modal contact time measured at the same point.

<sup>9</sup> CCR Title 22, Div. 4, Chap.3, Section 60301.230(a)(1)

<sup>10</sup> Or pursuant to CCR Title 22, Div. 4, Chap.3, Section 60301.230 (a)(2) A disinfection process that, when combined with the filtration process, has been demonstrated to inactivate and/or remove 99.999 percent of the plaque-forming units of F-specific bacteriophage MS2, or polio virus in the wastewater. A virus that is at least as resistant to disinfection as polio virus may be used for purposes of the demonstration.

2. Personnel involved in producing, transporting, or using recycled water shall be informed of possible health hazards that may result from contact and use of recycled water. <sup>T22, BPJ</sup>
3. Personnel involved in inspecting, maintaining or operating any distribution system equipment for recycled water shall be informed of the possible health hazards that may result from contact and use of recycled water. <sup>T22, BPJ</sup>
4. Delivery of recycled water shall cease during any period the Facility fails to produce "disinfected tertiary recycled water" meeting CCR Title 22 criteria. The delivery of recycled water shall not be resumed until all conditions which caused the limits to be violated have been corrected and effluent in the storage ponds is suitable for disinfected tertiary recycled water applications. <sup>BPJ</sup>
5. All recycled effluent storage reservoirs and use areas with public access shall be posted (in English and Spanish) to warn the public recycled wastewater is being stored or used. <sup>BPJ</sup>
6. Recycled water systems shall be properly labeled and regularly inspected to ensure proper operation, absence of leaks, and absence of illegal connections. <sup>BPJ, T22</sup>
7. Recycled effluent storage ponds and wastewater ponds shall have sufficient freeboard, no less than two feet<sup>11</sup> (measured vertically, from the water surface up to the point on the surrounding berm or dike having the lowest elevation and not including engineered outlet structures), at all times and shall be designed and constructed to prevent overtopping as a result of windy storm conditions. To determine pond freeboard, the Discharger shall install and maintain permanent markers with calibration indicating the water level at design capacity and available operational freeboard <sup>BPJ</sup>
8. The Supplier and Distributor shall maintain in good working order and operate as efficiently as possible any facility or control system installed by the Supplier, Distributor or Users to achieve compliance with this Order and all applicable water reclamation requirements.
9. The Supplier and Distributor shall implement, and ensure that Users implement, annual employee training to ensure proper operation of reclamation facilities, worker protection, and compliance with this Order.
10. The Supplier and Distributor shall ensure that all above-ground equipment, including pumps, piping, storage reservoir, and valves, etc. under their respective control which may at any time contain reclaimed water shall be adequately and clearly identified with warning signs. The Supplier and Distributor shall make all necessary provisions to inform the public that the water being stored or distributed is reclaimed municipal wastewater and is unfit for human consumption. The Supplier and Distributor shall ensure that each User complies with these requirements for all above-ground equipment under a User's control.
11. The Facilities shall be managed so as to minimize mosquito-breeding habitat. <sup>BPJ</sup>

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<sup>11</sup> Lesser freeboard, no less than one foot, is acceptable for below grade impoundments, and may be approved by the Executive Officer for above ground impoundments if documented by a registered civil engineer that structural integrity and required capacity will not be compromised with the proposed freeboard.

**Alarms**<sup>12</sup>

12. Alarm devices required for various unit processes as specified in other sections of these regulations shall be installed to provide warning of:
- a. Loss of power from the normal power supply.
  - b. Failure of a biological treatment process.
  - c. Failure of a disinfection process.
  - d. Failure of a coagulation process.
  - e. Failure of a filtration process.
  - f. Any other specific process failure for which warning is required by the regulatory agency.

All required alarm devices shall be independent of the normal power supply of the Facility.

13. The person to be warned shall be the plant operator, superintendent, or any other responsible person designated by the management of the reclamation plant and capable of taking prompt corrective action.
14. Individual alarm devices may be connected to a master alarm to sound at a location where it can be conveniently observed by the attendant. In case the reclamation plant is not attended full time, the alarm(s) shall be connected to sound at a police station, fire station or other full-time service unit with which arrangements have been made to alert the person in charge at times that the reclamation plant is unattended.

**Power Supply**<sup>13</sup>

15. The power supply shall be provided with one of the following reliability features:
- a. Alarm and standby power source.
  - b. Alarm and automatically actuated short-term retention or disposal provisions as specified in Title 22 Section 60341.
  - c. Automatically actuated long-term storage or disposal provisions as specified in Title 22 Section 60341.

**Flexibility of Design**<sup>14</sup>

16. The design of process piping, equipment arrangement, and unit structures in the reclamation plant must allow for efficiency and convenience in operation and maintenance and provide flexibility of operation to permit the highest possible degree of treatment to be obtained under varying circumstances.

**Personnel**<sup>15</sup>

17. Each reclamation plant shall be provided with a sufficient number of qualified personnel to operate the facility effectively so as to achieve the required level of treatment at all times.
18. Qualified personnel shall be those meeting requirements established pursuant to Chapter 9 (commencing with Section 13625) of the Water Code.

**Maintenance**<sup>16</sup>

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<sup>12</sup> CCR Title 22, Div. 4, Chap. 3, Section 60335

<sup>13</sup> CCR Title 22, Div. 4, Chap. 3, Section 60337

<sup>14</sup> CCR Title 22, Div. 4, Chap. 3, Section 60333

<sup>15</sup> CCR Title 22, Div. 4, Chap. 3, Section 60325

<sup>16</sup> CCR Title 22, Div. 4, Chap. 3, Section 60327

19. A preventive maintenance program shall be provided at each reclamation plant to ensure that all equipment is kept in a reliable operating condition.

**Operating Records and Reports**<sup>17</sup>

20. Operating records shall be maintained at the reclamation plant or a central depository within the operating agency. These shall include: all analyses specified in the reclamation criteria; records of operational problems, plant and equipment breakdowns, and diversions to emergency storage or disposal; all corrective or preventive action taken.
21. Process or equipment failures triggering an alarm shall be recorded and maintained as a separate record file. The recorded information shall include the time and cause of failure and corrective action taken.
22. A monthly summary of operating records as specified in these requirements shall be filed monthly with the regulatory agency.<sup>18</sup>
23. Any discharge of untreated or partially treated wastewater to the use area, and the cessation of same, shall be reported immediately by telephone to Water Board staff, the State Department of Health, and the local health officer at the numbers provided in the Monitoring and Reporting Requirements.

**Bypass**<sup>19</sup>

24. There shall be no bypass of untreated or partially treated wastewater from the reclamation plant or any intermediate unit processes to the point of use.

**Off-Specification Effluent Contingency Plan**

25. In the event effluent discharged to the effluent storage ponds does not meet the criteria for disinfected tertiary recycled water, the Supplier shall implement the Off-Specification Contingency Plan.<sup>20, ROWD</sup>
26. The Off-Specification Contingency Plan shall be reviewed and updated annually as necessary. A copy of the revised Off-Specification Contingency Plan or statement indicating the Plan has been reviewed, but not updated shall be submitted to the Water Board as part of the annual monitoring report.<sup>BPJ</sup>
27. Alternative reuse methods for off-specification effluent may be implemented on an as needed basis if they meet the criteria for the "Uses of Recycled Water" contained in CCR Title 22, Division 4, Chapter 3, Article 3 (Sections 60303-60309) and prior approval is given by the Water Board and DPH.<sup>BPJ</sup>

**Sludge and Solid Waste**

(Sludge in this document means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screening material generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment. Biosolids refers to sludge that has been

<sup>17</sup> CCR Title 22, Div. 4, Chap. 3, Section 60329

<sup>18</sup> Per CCR Title 22 Div. 4, Chap. 3, Section 60301.740. "Regulatory agency" means the California Regional Water Quality Control Board(s) that have jurisdiction over the recycling plant and use areas.

<sup>19</sup> CCR Title 22, Div. 4, Chap. 3, Section 60331

<sup>20</sup> August 2005, RMC, Proposed Pond #1 Monitoring Plan

treated and tested and shown to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation activities.)

28. Sludge and solid waste shall be removed from treatment facilities as needed to ensure optimal plant operation.
29. Treatment and storage of sludge shall be confined on-site and conducted in a manner that precludes infiltration of waste constituents into soils in a mass or concentration that will violate Groundwater Limitations.
30. Any storage of residual sludge and solid waste shall be temporary and controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or concentration that will violate Groundwater Limitations.
31. Sludge and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment, disposal, or reuse at sites (i.e., landfill, composting sites, soil amendment sites) operated in accordance with valid waste discharge requirements issued by a regional water quality control board will satisfy this specification.
32. Use of biosolids as a soil amendment shall comply with valid waste discharge requirements issued by a regional water quality control board. In most cases, this will mean the General Biosolids Order (SWRCB Water Quality Order No. 2004-0012-DWQ, General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities). For a biosolids use project to be covered by the General Biosolids Order, the Discharger must file a complete Notice of Intent and receive a Notice of Applicability for each project.
33. Use and disposal of biosolids should comply with the self-implementing federal regulations of Title 40, Code of Federal Regulations (CFR), Part 503, which are subject to enforcement by the U.S. Environmental Protection Agency not the Water Board. If during the life of this Order the State accepts primacy for implementation of 40 CFR 503, the Water Board may also initiate enforcement where appropriate.

#### **General Requirements**

34. Extraneous surface drainage shall be excluded from the wastewater treatment and effluent storage facilities. <sup>BPJ</sup>
35. Best management practices shall be implemented to minimize the inflow and infiltration of storm water and/or unauthorized wastewater into the Facility. <sup>BPJ</sup>
36. All storm water contacting raw domestic wastewater or disinfected tertiary recycled water shall be contained and managed as such. <sup>BPJ</sup>
37. The Supplier shall provide weekly irrigation reports to the Distributor and Users documenting Facility influent flows, User irrigation flows (including Facility irrigation flows reported separately), and the amount of recycled effluent in storage and remaining storage capacity. <sup>BPJ</sup>

**D. USER REQUIREMENTS<sup>21</sup>**

1. The application of disinfected tertiary recycled water is limited to the following areas pursuant to Title 22, Division 4, Chapter 3, Section 60304 of the California Code of Regulations:
  - a. Food crops, including all edible root crops, where the recycled water comes into contact with the edible portion of the crop,
  - b. Parks and playgrounds,
  - c. School yards,
  - d. Residential landscaping,
  - e. Unrestricted access golf courses,<sup>22</sup> and
  - f. Any other irrigation use not specified in Section 60304 (Title 22) and not prohibited by other sections of the California Code of Regulations, or within these requirements.
2. The Supplier and Distributor may add additional use areas/Users for the application of disinfected tertiary treated wastewater as long as they meet all applicable requirements contained within this Order and the California Code of Regulations.
3. No irrigation with disinfected tertiary recycled water shall take place within 50 feet of any domestic water supply well unless all of the following conditions have been met:
  - a. A geological investigation demonstrates that an aquitard exists at the well between the uppermost aquifer being drawn from and the ground surface.
  - b. The well contains an annular seal that extends from the surface into the aquitard.
  - c. The well is housed to prevent any recycled water spray from coming into contact with the wellhead facilities.
  - d. The ground surface immediately around the wellhead is contoured to allow surface water to drain away from the well.
  - e. The owner of the well approves of the elimination of the buffer zone requirement.
4. No impoundment of disinfected tertiary recycled water shall occur within 100 feet of any domestic water supply well.
5. Any use of recycled water shall comply with the following:
  - a. Any irrigation runoff shall be confined to the recycled water use area, unless the runoff does not pose a public health threat and is authorized by the regulatory agency.
  - b. Spray, mist, or runoff shall not enter dwellings, designated outdoor eating areas, or food handling facilities.
6. Drinking water fountains shall be protected against contact with recycled water spray, mist, or runoff.
7. Spray irrigation of recycled water shall be accomplished at a time and in a manner to minimize ponding and the possibility of public contact with sprayed materials.<sup>BPJ</sup>

<sup>21</sup> CCR Title 22, Div. 4, Chap. 3, Section 60310

<sup>22</sup> For golf course use, the scorecards must clearly state that reclaimed water is used for irrigation. <sup>BPJ</sup>

8. No spray irrigation of any recycled water, other than disinfected tertiary recycled water, shall take place within 100 feet of a residence or a place where public exposure could be similar to that of a park, playground, or school yard.
9. All use areas where recycled water is used that are accessible to the public shall be posted with signs that are visible to the public, in a size no less than 4 inches high by 8 inches wide, that include the following wording: "RECYCLED WATER - DO NOT DRINK". Each sign shall display an international symbol similar to that shown in figure 60310-A of CCR Title 22, Section 60310. The Department may accept alternative signage and wording, or an educational program, provided the applicant demonstrates to the Department that the alternative approach will assure an equivalent degree of public notification.
10. Except as allowed under section 7604 of title 17, California Code of Regulations, no physical connection shall be made or allowed to exist between any recycled water system and any separate system conveying potable water.
11. The portions of the recycled water piping system that are in areas subject to access by the general public shall not include any hose bibs. Only quick couplers that differ from those used on the potable water system shall be used on the portions of the recycled water piping system in areas subject to public access.
12. The Distributor shall ensure that backflow prevention devices are in proper working order by testing initially and annually thereafter, in accordance with CCR Title 18, Section 7605. Reports of testing and maintenance shall be maintained by the Distributor.

### **Design Requirements**

13. The public water supply shall not be used as a backup or supplemental source of water for a dual-plumbed recycled water system unless the connection between the two systems is protected by an air gap separation which complies with the requirements of sections 7602(a) and 7603(a) of title 17, California Code of Regulations, and the approval of the public water system has been obtained.<sup>23</sup>
14. All pipes installed above or below the ground, on and after June 1, 1993, that are designed to carry recycled water, shall be colored purple or distinctively wrapped with purple tape.<sup>24</sup>
15. The Distributor shall implement a Cross Connection Control Plan<sup>25</sup> to protect the public water supply system. The Cross Connection Plan shall be reviewed and updated annually as necessary. A copy of the revised Plan or statement indicating the Plan has been reviewed, but not updated shall be submitted to the Water Board as part of the Distributor's annual monitoring report. ROWD, BPJ

### **Nutrient Management Plan**

16. Hydraulic and nutrient loading rates for the application of disinfected tertiary recycled water shall be based on food crop, vegetation or landscaping consumption and tolerance and shall not exceed what is reasonable for production of the food crops, vegetation or landscaping (i.e., recycled water shall be applied in an amount that will not cause nitrogen within the root

<sup>23</sup> CCR Title 22, Div. 4, Chap. 3, Section 60315

<sup>24</sup> California Health & Safety Code Section 116815

<sup>25</sup> August 2005, RMC, Engineering Report for Production, Distribution, and use of Recycled Water, Appendix E – Las Palmas Ranch Cross Connection Control Plan

zone to exceed the agronomic demand for nitrogen and result in the leaching of nitrate to groundwater).<sup>BPJ</sup>

17. The Supplier and Distributor shall prepare and implement a nutrient management plan for the application of recycled water to protect the beneficial uses of groundwater. The plan shall account for all nutrient loading to the application areas and ensure that the total amount of nitrogen applied does not exceed the amount of nitrogen required by the food crops, vegetation or landscaping being irrigated.
18. As part of the nutrient management plan, the Supplier and Distributor shall submit an annual report documenting allowable and actual nitrogen loading to the recycled water application areas. The report shall include, at a minimum:
  - a. Analysis of the contributing sources of nutrients being applied to the recycled water application areas;
  - b. Analysis of annual nitrogen loading to the basin and individual application areas from each contributing source;
  - c. Analysis of the allowable nutrient and hydraulic loading (based on limiting nitrogen loading) of recycled water based on characteristic effluent data for nitrogen, other contributing nitrogen sources, and the nutritive requirements of the application areas;
  - d. Comparison of the actual and allowable annual nitrogen loading rates;
  - e. Analysis of groundwater monitoring data for nitrogen constituents;
  - f. Evaluation of potential impacts of nutrient loading on the groundwater basin;
  - g. Evaluation of potential nutrient reduction measures; and,
  - h. Recommendations and time schedules for the implementation of measures addressing excessive nitrogen loading (i.e. actual loading greater than allowable loading) as applicable.
19. **Annual nutrient management reports are due January 31<sup>st</sup> of each year** and may be included as part of the annual monitoring report. **The first annual nutrient management plan report is due January 31, 2009.** The plan shall be reviewed and updated annually thereafter as necessary. A copy of the revised plan or statement indicating the plan has been reviewed, but not updated shall be submitted to the Water Board as part of the annual monitoring reports.
20. Additional annual nutrient management reports will not be required upon request by the Supplier and Distributor and approval by the Executive Officer given the following conditions are met:
  - a. The initial nitrogen loading evaluation indicates the application of recycled water at appropriate hydraulic rates along with other nitrogen sources will not exceed the nutritive requirements of the food crops, vegetation or landscaping being irrigated
  - b. Recycled water is not over applied in an effort to increase disposal that may result in significant soil flushing and runoff;
  - c. A nutrient management plan is implemented for the controlled application of fertilizers by landscaping contractors maintaining the application areas; and,
  - d. Effluent nitrogen concentrations from the Facility regularly meet or are less than the effluent limitations of this Order and are stable.

(Approval of this variance is contingent on reasonable and scientifically defensible assumptions being applied to the loading evaluation.)

21. Discharges that exceed the hydraulic loading rate based on the nutritive requirements of the receiving vegetation may be allowable on a case-by-case basis upon request by the Distributor and approval by the Executive Officer given the following conditions are met:
- a. The nitrogen loading evaluation indicates the land application of wastewater at appropriate hydraulic rates (based on soil permeability) will not exceed the nutritive requirements of the vegetation being irrigated by more than a total nitrogen concentration as determined by the following equation<sup>26</sup>:

$$AN = (TOC - 5) / 2$$

TOC = effluent Total Organic Carbon

- b. Wastewater is not over applied in an effort to increase disposal that may result in significant soil flushing and runoff;
- c. Effluent nitrogen concentrations from the Facility regularly meet or are less than the effluent limitations of this Order and are stable; and,
- d. The Discharger provides an assimilative capacity analysis and nitrogen balance showing that the additional nutrient loading to the groundwater basin will not cause or contribute to exceedances of water quality objectives for nitrate in groundwater

(Approval of this variance is contingent on reasonable and scientifically defensible assumptions being applied to the assimilative capacity analysis and nitrogen balance.)

### **Salts Management Program**

22. The Supplier and Distributor shall implement a salts management program to document salt loading and evaluate and implement measures for the reduction of salt loading as the result of the application of recycled water. Salt reduction measures shall focus on all potential salt contributions from the water supply, and residential, commercial and industrial uses as applicable prior to disposal. The Supplier and Distributor shall evaluate limiting or prohibiting domestic water softeners and conditioners under California Health and Safety Code Section 116786 and shall adopt an ordinance under Section 116786 as appropriate and feasible to reduce salt loading from the domestic use of water softeners.
23. As part of the salts management program, the Supplier and Distributor shall submit an annual report documenting salt loading and salt reduction efforts. This report shall include, at a minimum:
- a. Analysis of annual salt (TDS, sodium, chloride, sulfate, and boron) loading to the basin and individual application areas;
  - b. Analysis of the contributing sources of salt mass in the recycled water (including the evaporative concentration of salts within the effluent storage ponds);
  - c. Analysis of groundwater monitoring data for salt constituents;
  - d. Evaluation of potential impacts of salt loading on the groundwater basin;
  - e. Evaluation of potential salt reduction measures including a water softener ordinance;
  - f. Summary of existing salt reduction measures and their impact; and,

<sup>26</sup> Maximum of nitrogen that can be effectively denitrified during rapid infiltration under optimum operating conditions; Metcalf and Eddy, Third Ed., 1991, page 972.

- g. Recommendations and time schedules for implementation of proposed salt reduction measures.

29. Annual salts management reports are due January 31<sup>st</sup> of each year and may be included as part of the annual monitoring report. The first annual salts management report is due January 31, 2009.

#### Groundwater Limitations

24. The discharge shall not cause the pH of underlying groundwater to exceed 8.3 or recede below 6.5. <sup>BP</sup>
25. The use or disposal of treated wastewater shall not cause the median concentration of coliform organisms in groundwater over any seven-day period to be more than 2.2/100 ml. <sup>BP, BPJ</sup>
26. The use or disposal of treated wastewater shall not cause a statistically significant increase of mineral or organic constituent concentrations in underlying groundwater, as determined by statistical analysis of samples collected from wells in the vicinity of the disposal area. <sup>BP, BPJ</sup>
27. The use or disposal of treated wastewater shall not cause nitrate concentrations in affected groundwater to exceed 8 mg/l (as N) and shall not cause a statistically significant increase of nitrate concentrations in underlying groundwater. <sup>27, BPJ, BP</sup>
28. The use or disposal of treated wastewater shall not cause groundwater to contain taste- or odor-producing substances in concentrations that adversely affect beneficial uses. <sup>BP</sup>
29. To protect the *municipal and domestic supply* beneficial uses of groundwater underlying the use or disposal areas, the application of treated wastewater shall not cause groundwater to: <sup>BP, BPJ, T22</sup>
- a. exceed the Primary Maximum Contaminant Levels for organic chemicals set forth in the California Code of Regulations, Title 22, Division 4, Chapter 15, Article 5.5, Section 64444.
  - b. exceed the Primary Maximum Contaminant Levels for inorganic chemicals set forth in the California Code of Regulations, Title 22, Division 4, Chapter 15, Article 4, Section 64431.
  - c. exceed the levels for radionuclides set forth in the California Code of Regulations, Title 22, Division 4, Chapter 15, Article 5, Section 64443.
30. The use or disposal of treated wastewater shall not cause radionuclides to be present in groundwater in concentrations that are deleterious to human, plant, animal, or aquatic life, or result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life. <sup>BP</sup>

#### Individual Recycled Water Use Permits

<sup>27</sup> The evaluation of this requirement will consider pre-existing conditions based on available characteristic groundwater quality data in the vicinity of the use areas.

30. The Supplier and Distributor shall enforce rules and regulations for recycled water users governing the design, construction and maintenance of recycled water use facilities and the use of recycled water, in accordance with the uniform statewide reclamation criteria established pursuant to California Water Code Section 13521.<sup>28</sup> The Supplier and Distributor shall also develop administrative procedures specifying how the recycled water rules and regulations and permit-based system for regulating users will be implemented. **The implementation procedures shall be submitted to the Department of Public Health and Central Coast Water Board by October 31, 2008, for review and approval.** The rules and regulations shall be reviewed and updated annually thereafter as necessary. A copy of the revised rules and regulations or statement indicating they have been reviewed, but not updated shall be submitted to the Water Board as part of the annual monitoring reports.
31. The Supplier and Distributor shall require each User to (i) designate a Reclaimed Water Site Supervisor responsible for compliance with permit conditions and answerable to the Supplier and Distributor<sup>29</sup>, and (ii) immediately notify the Supplier and Distributor of changes in the Reclaimed Water Site Supervisor and provide documentation that the new supervisor has received training.
32. Recycled Water Use permits, issued by the Supplier and Distributor in accordance with the approved rules and regulations, form the basis of permitted recycled water use by specific Users. Recycled Water Use permits shall specify self-monitoring and reporting requirements for each User, and require compliance with all applicable requirements of this Order. The Distributor must provide a copy of the Recycled Water Use permit and this Order to the Users. Recycled Water Use permits shall require Users to have these available at all times for inspection by Water Board staff, the Distributor, or State/County Health Officers.
33. If someone other than the User is responsible for applying the recycled water (i.e. secondary distributor like a truck hauler) then the Supplier and Distributor shall inform the secondary distributor of these requirements in a written permit or other suitable manner. In addition, the secondary distributor shall fill out a Recycled Water Release Form when receiving reclaimed water from the Supplier and Distributor. The secondary distributors must carry the Recycled Water Release Form at all times.

## E. PROVISIONS

1. This Order supersedes Order No. R3-2005-0074 for all uses specified by that Order. Order No. R3-2005-0074 is hereby rescinded.
2. The Supplier and Distributor shall comply with all applicable requirements of Monitoring and Reporting Program No. R3-2008-0042 as adopted by the Water Board and as may be amended by the Executive Officer. The Supplier and Distributor shall be responsible for collecting necessary data and reports from the Users. The Supplier and Distributor shall require Users to appoint and train a Reclaimed Water Supervisor and to submit on-site observation reports and use data to the Supplier and Distributor, who will compile and file self-monitoring reports with the Water Board. The Supplier and Distributor, at its discretion, may appoint and train the Users' Reclaimed Water Supervisors and collect on-site observation reports and use data.

<sup>28</sup> CWC Section 13523.1(b)(3)

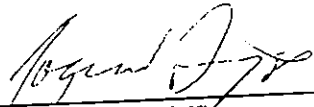
<sup>29</sup> CCR Title 17, Division 1, Chapter 5, Group 4, Article 1, Section 7586

3. The Supplier and Distributor shall develop a groundwater monitoring plan capable of determining the impact of treated wastewater and recycled water upon underlying groundwaters. The Groundwater Monitoring Plan shall be submitted to the Department of Public Health and Water Board by October 31, 2008, for review and approval.
4. The Supplier shall be responsible for ensuring and documenting that reclaimed water meets the quality standards of this Order. The Distributor shall be responsible for regulating the design, construction, maintenance and operation of recycled water transport facilities, application areas and associated appurtenances owned and operated by the Users and for ensuring that Users meet all water application, operations and maintenance requirements of this Order. The Distributor shall conduct periodic inspections of User facilities and conduct monitoring and reporting to document compliance with the conditions of the Users' permits and this Order.
5. The Supplier, Distributor and Users shall permit the Water Board staff or its authorized representative in accordance with California Water code section 13267(c):
  - Entry upon premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of the Order,
  - Access to and copy of any records that must be kept under conditions of this Order,
  - Inspection of any facility, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order,
  - To photograph, sample, and monitor for the purpose of assuring compliance with this Order.
6. Prior to the initial delivery of recycled water to each use site, the Producer shall submit piping plans for that site to California Department of Public Health for approval.
7. For any extension or expansion of the recycled water system or use areas not covered by the Title 22 report, the Producer shall submit to California Department of Public Health an addendum to the Title 22 report for approval.
8. Upon Executive Officer approval, additional storage/percolation may be allowed at the scalping plant site.
9. The Supplier and Distributor shall comply with all applicable items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated January 1984. The Water Board will revise this Order periodically and may revise these requirements when necessary.
10. Pursuant to CCR Title 23, Division 3, Chapter 9, , the Discharger must submit a written report to the Executive Officer not later than January 31, 2018, addressing:
  - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
  - b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.

July 11, 2008

I, **ROGER W. BRIGGS, Executive Officer**, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on July 11, 2008

Ordered By:

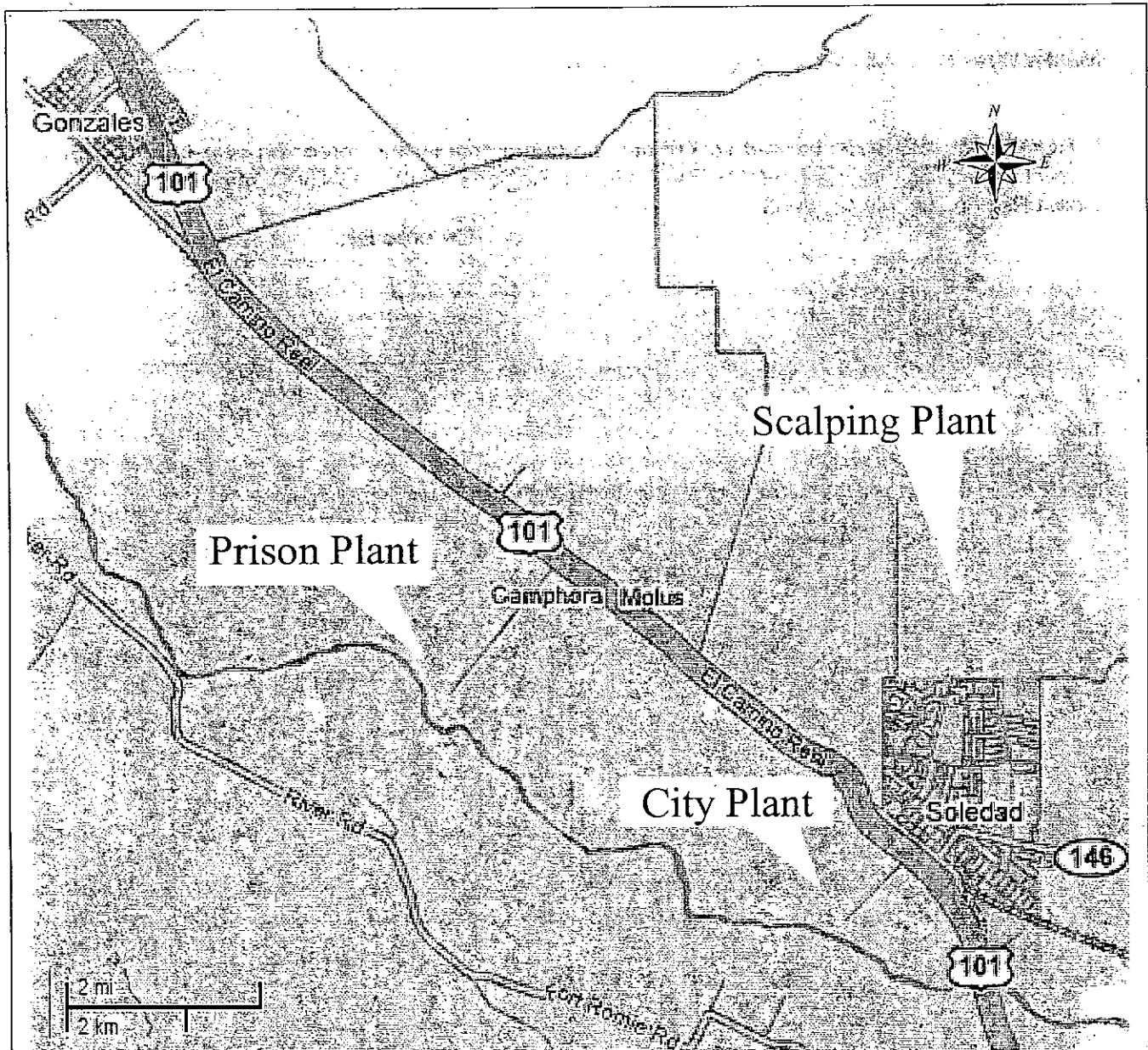
  
Executive Officer

TJK

126-01

Paper File: City of Soledad

Electronic File: S:\WDR\WDR Facilities\Monterey Co\City of Soledad WWTP\Reclamation Requirements\WRR.doc



Location Map  
City of Soledad  
Recycled Water Facilities  
Monterey County, California

Attachment

A



**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL COAST REGION  
895 Aerovista Place, Suite 101  
San Luis Obispo, CA 93401-7906**

**MONITORING AND REPORTING PROGRAM REQUIREMENTS  
ORDER NO. R3-2008-0042**

**(Waste Discharger Identification No. 3 270113001)**

**FOR  
THE CITY OF SOLEDAD  
WATER RECYCLING FACILITY  
MONTEREY COUNTY**

This Monitoring and Reporting Program Requirements Order (MRP) is issued by the Regional Water Quality Control Board, Central Coast Region (Water Board) pursuant to California Water Code section 13267 and is incorporated into Master Reclamation Permit Order No. R3-2008-0042. The City of Soledad is subject to the MRP because it discharges recycled water. This MRP is necessary to assure that the discharge of recycled water complies with the requirements of the Master Reclamation Permit and are protective of public health and the environment.

**SUPPLIER REQUIREMENTS**

**A. WATER SUPPLY MONITORING**

1. The Supplier shall submit water quality data representing the quality of the supply water used by the City of Soledad and the California Department of Correction Prisons located in Soledad. The data shall include all chemicals and parameters that are required to be part of a Health & Safety Code §116470 Consumer Confidence Report.

**B. INFLUENT MONITORING**

1. Representative samples of the treatment facility influent shall be collected and analyzed for the constituents/parameters and at the frequencies specified in the following table:

| Constituent/Parameter <sup>a</sup> | Units | Sample Type <sup>b</sup> | Sampling Frequency <sup>c</sup> |
|------------------------------------|-------|--------------------------|---------------------------------|
| pH                                 | Units | Grab                     | Weekly                          |
| BOD <sub>5</sub>                   | mg/L  | 24-hour composite        | Weekly                          |
| Settleable Solids                  | mg/l  | 24-hour composite        | Weekly                          |
| Total Suspended Solids             | mg/L  | 24-hour composite        | Weekly                          |
| Total Nitrogen (as N)              | mg/L  | Grab                     | Monthly                         |
| Nitrate (as Nitrogen)              | mg/l  | Grab                     | Monthly                         |
| Ammonia (as N)                     | mg/L  | Grab                     | Monthly                         |
| Sulfate                            | mg/L  | Grab                     | Quarterly                       |
| Boron                              | mg/L  | Grab                     | Quarterly                       |
| Total Dissolved Solids             | mg/L  | Grab                     | Quarterly                       |
| Sodium                             | mg/L  | Grab                     | Quarterly                       |
| Chloride                           | mg/L  | Grab                     | Quarterly                       |

Notes:

- a. Sampling for specific analytes may be reduced or discontinued after one year upon Discharger request and Executive Officer approval for parameters/constituents for which additional data provides no benefit.
- b. Composite samples shall be flow weighted. 24-hour composite samples shall be collected on a Monday through Sunday rotating schedule and subsequent monthly sampling events shall be separated by at least 16 days and no greater than 30 days.
- c. Quarterly monitoring shall be conducted in January, April, July, and October.

### C. EFFLUENT MONITORING

1. Representative samples of treatment facility effluent shall be collected<sup>a</sup> and analyzed for the constituents/parameters and at the frequency specified below for each treatment facility:

| Constituent/Parameter <sup>b</sup>      | Units      | Sample Type <sup>c</sup>       | Sampling Frequency     |
|-----------------------------------------|------------|--------------------------------|------------------------|
| Min Daily Flow                          | gal/day    | Metered                        | Continuous             |
| Max Daily Flow                          | gal/day    | Metered                        | Continuous             |
| Average Daily Flow                      | gal/day    | Calculated                     | Daily                  |
| Turbidity                               | NTU        | Metered                        | Continuous             |
| Total Chlorine Residual <sup>d, e</sup> | mg/L       | Metered                        | Continuous             |
| Coliform                                | MPN/100 ml | Grab                           | Daily                  |
| BOD <sub>5</sub>                        | mg/L       | 24-hour composite <sup>c</sup> | Weekly <sup>f</sup>    |
| Total Suspended Solids                  | mg/L       | 24-hour composite <sup>c</sup> | Weekly <sup>f</sup>    |
| pH                                      | Units      | Grab                           | Weekly <sup>f</sup>    |
| Total Nitrogen (as N)                   | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Nitrate (as N)                          | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Nitrite (as Nitrogen)                   | mg/l       | Grab                           | Quarterly <sup>g</sup> |
| Ammonia (as N)                          | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Total Kjeldahl Nitrogen (as N)          | mg/l       | Grab                           | Quarterly <sup>g</sup> |
| Total Dissolved Solids                  | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Sodium                                  | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Chloride                                | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Sulfate                                 | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Boron                                   | mg/L       | Grab                           | Quarterly <sup>g</sup> |
| Aluminum                                | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Antimony                                | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Arsenic                                 | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Barium                                  | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Beryllium                               | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Cadmium                                 | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Chromium                                | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Copper                                  | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Cyanide                                 | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Flouride                                | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Lead                                    | mg/l       | Grab                           | Annually <sup>h</sup>  |
| Mercury                                 | mg/l       | Grab                           | Annually <sup>h</sup>  |

|            |      |      |                           |
|------------|------|------|---------------------------|
| Nickel     | mg/l | Grab | Annually <sup>h</sup>     |
| Selenium   | mg/l | Grab | Annually <sup>h</sup>     |
| Thalium    | mg/l | Grab | Annually <sup>h</sup>     |
| Zinc       | mg/l | Grab | Once/5 Years <sup>i</sup> |
| VOCs       | mg/l | Grab | Once/5 Years <sup>i</sup> |
| PCBs       | mg/l | Grab | Once/5 Years <sup>i</sup> |
| Pesticides |      |      |                           |

Notes:

- Sampling shall occur immediately following the final treatment process (i.e. disinfection or dechlorination as applicable) unless noted otherwise
- Sampling for specific analytes may be reduced or discontinued after one year upon Discharger request and Executive Officer approval for parameters/constituents for which additional data provides no benefit.
- Composite samples shall be flow weighted. 24-hour composite samples shall be collected on a Monday through Sunday rotating schedule and subsequent monthly sampling events shall be separated by at least 16 days and no greater than 30 days.
- Shall be compared to the chlorine residual required to achieve a minimum CT value of 450 milligram-minutes per liter.
- Chlorine residual monitoring is not required for ultraviolet (UV) disinfection. The Executive Officer may require alternative sampling and reporting requirements for UV disinfection based on manufacturer and DHS recommendations to achieve minimum disinfection.
- Weekly samples shall be collected on a Monday through Sunday rotating schedule. Subsequent monthly sampling events shall be separated by at least 16 days and no greater than 30 days.
- Quarterly monitoring shall be conducted in January, April, July, and October
- Annual monitoring shall be conducted in July
- Once/5 Years monitoring shall commence in July 2009

#### D. STORAGE FACILITY MONITORING

Recycled water storage areas shall be inspected daily. Weekly visual inspections will be allowable for storage reservoirs fitted with reliable electronic remote depth gauging systems. Notes shall be kept of observations and shall be summarized in annual monitoring reports. In the event of impending freeboard violation, storage pond overflow, or backflow into the treatment plant, the Distributor (and Users as appropriate), Executive Officer, and the Department of Health Services shall be notified immediately.

#### E. SOLIDS/BIOSOLIDS MONITORING

- The following information shall be submitted with the Annual Report required by Standard Provision C.16:

- Annual biosolids removed in dry tons and percent solids.
- If appropriate, a narrative description of biosolids dewatering and other treatment processes, including process parameters. For example, if drying beds are used, report depth of application and drying time. If composting is used, report the temperature achieved and duration.
- A description of disposal methods, including the following information as applicable related to the disposal methods used at the facility. If more than one method is used,

include the percentage and tonnage of annual biosolids production disposed by each method.

- i. For landfill disposal include: 1) the Water Board WDR numbers that regulate the landfills used, 2) the present classifications of the landfills used, and 3) the names and locations of the facilities receiving biosolids.
- ii. For land application include: 1) the location of the site(s), 2) the Water Board's WDR numbers that regulate the site(s), 3) the application rate in lbs/acre/year (specify wet or dry), and 4) subsequent uses of the land.
- iii. For offsite application by a licensed hauler and composter include: 1) the name, address and USEPA license number of the hauler and composter.
- d. Copies of analytical data required by other agencies (i.e. USEPA or County Health Department) and licensed disposal facilities (i.e. landfill, land application, or composting facility) for the previous year.

### **DISTRIBUTOR REQUIREMENTS**

#### **F. RECYCLED WATER USE AREA MONITORING**

1. The quantity of reclaimed water distributed to each reuse site shall be recorded on a weekly basis. Total flows shall be metered or estimated based on irrigation run times and distribution system design flow rates. Total as applied flows shall be compared to Supplier effluent flow rates.
2. During periods of recycled water application the Distributor or Users, as applicable, shall inspect the irrigation use areas no less frequently than weekly to verify and document compliance with Order No R3-2008-0042. The visual inspections shall be noted in a bound inspection logbook(s) and at a minimum shall document proper sprinkler operation, runoff, erosion, saturated surface conditions, and odors. The logbook(s) shall be made available to the Water Board and California Department of Public Health upon request. A summary of observations made during water recycling area inspections and a brief discussion of any corrective actions taken or planned shall be included with each annual monitoring report.
3. The Distributor and Users shall coordinate with the California Department of Public Health to ensure and document that backflow devices are present, tested annually by a certified individual, and repaired or replaced if found defective.
4. The Distributor shall inspect and document the operation of the reuse site irrigation systems at least quarterly to verify that the Users are operating the reuse sites in compliance with the uniform statewide reclamation criteria established pursuant to California Water Code Section 13521 and Order No. R3-2006-0041<sup>1</sup>.
5. The Distributor in coordination with the Users, water purveyor, and California Department of Public Health shall perform and document a cross-connection test by an appropriately certified individual on an annual basis at each reuse site where both recycled water and potable water piping systems are utilized for irrigation or are otherwise present in proximity to each other.<sup>2</sup>

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<sup>2</sup> Cross-connection tests will not be required for portions of the distribution system or reuse site areas for which no distribution system or potable water system maintenance, modifications, or additions have occurred since the last cross-

6. Each individual User Reclaimed Water Site Supervisor shall provide quarterly updates to the Distributor regarding irrigation frequency and flow rates, proposed system modifications, system peculiarities, and to verify employee training. The Distributor shall keep a record of all system modifications and document that all work is conducted in accordance with the Cross Connection Control Plan and applicable regulations.
7. The Distributor shall compile and conduct quarterly reviews of the applied recycled water flows to identify unusual usage behavior or significant changes. The Distributor shall conduct and document follow-up investigations if patterns change dramatically.

## G. GROUNDWATER MONITORING

1. Representative samples of groundwater shall be collected from shallow wells<sup>a</sup> upgradient and downgradient of disposal areas. To ascertain compliance with waste discharge requirements in establishing new, or verifying existing upgradient and downgradient monitoring wells, the monitoring network shall be supported by sufficient, as determined by the Executive Officer, geologic and hydrogeologic documentation. Samples of groundwater shall be collected and analyzed for the constituents and at the frequencies specified in the following table:

| Constituent/Parameter <sup>b</sup> | Units | Sample Type | Sampling Frequency     |
|------------------------------------|-------|-------------|------------------------|
| Depth to Water                     | Feet  | Measured    | Quarterly <sup>c</sup> |
| Total Nitrogen (as N)              | mg/L  | Grab        | Quarterly              |
| Total Kjeldahl Nitrogen (as N)     | mg/L  | Grab        | Quarterly              |
| Nitrate (as N)                     | mg/L  | Grab        | Quarterly              |
| Nitrite (as N)                     | mg/L  | Grab        | Quarterly              |
| pH                                 | —     | Grab        | Quarterly              |
| Total Dissolved Solids             | mg/L  | Grab        | Quarterly              |
| Sodium                             | mg/L  | Grab        | Quarterly              |
| Chloride                           | mg/L  | Grab        | Quarterly              |
| Sulfate                            | mg/L  | Grab        | Quarterly              |
| Boron                              | mg/L  | Grab        | Quarterly              |

### Notes:

- a) Provide well construction details for each monitoring well indicating total well depth and screen interval in depth below ground surface and elevations referenced to MSL along with the top of well casing elevation
- b) Sampling for specific analytes or from specific monitoring wells may be reduced or discontinued after one year upon Discharger request and Executive Officer approval for parameters/constituents for which additional data provides no benefit.
- c) Quarterly monitoring shall be conducted in January, April, July, and October

Based on review of available groundwater data the Executive Officer may request geologic and hydrogeologic documentation to ascertain whether the existing monitoring well network is sufficient to verify compliance with waste discharge requirements and whether additional monitoring wells are required

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connection test. The Distributor shall provide a certified statement as such for portions of the distribution system or reuse sites not tested for potential cross-connection.

## H. REPORTING

1. The Distributor shall submit **Quarterly** monitoring reports summarizing reclaimed water use at each reuse site and shall including a list of the reuse sites with the name, location and brief description of each reuse site, the total amount of reclaimed water supplied to each reuse site, and the name of the hydrologic areas underlying each reuse site [Required pursuant to California Water Code Section 13523.1(b)(5)]. The Quarterly monitoring reports shall be submitted as follows:

| Monitoring Occurring in... | Shall be reported by...  |
|----------------------------|--------------------------|
| January 1 – March 31       | April 30 <sup>th</sup>   |
| April 1 – June 30          | July 31 <sup>st</sup>    |
| July 1 – September 30      | October 31 <sup>st</sup> |
| October 1 – December 31    | January 31 <sup>st</sup> |

2. The Distributor shall submit annual reports by January 31<sup>st</sup> of each year summarizing reclaimed water use, including the total volume of reclaimed water supplied, and the total number of reclaimed use sites and their locations. Reports shall include records of the Distributor's or User's reuse site inspections and results of the annual cross-connection tests. Annual monitoring reports shall contain at a minimum:
  - a. Documentation of the use area and groundwater monitoring requirements listed above;
  - b. A list of Users accompanied by a scaled map showing the recycled water use areas;
  - c. A table summarizing monthly recycled water application rates in acre-feet and gallons for each recycled water use;
  - d. The Nutrient Management Plan report as specified in Distributor/User Requirements;
  - e. The Salt Management Program report as specified in Distributor/User Requirements; and,
  - f. An up-to-date copy of the Rules and Regulations for Recycled Water Users and a standard copy of the Recycled Water Use Permit issued to the Users as specified in Individual Recycled Water Use Permits.
3. The monitoring reports shall contain all data collected or calculated over the previous annual monitoring period. All monitoring data shall be tabulated in a logical and coherent format and be accompanied by copies of laboratory analytical data sheets as applicable. The data shall be summarized in a manner that clearly illustrates compliance with the Order. The User list, use area map, and recycled water application summary table shall be cross-referenced for easy evaluation.
4. The Distributor shall report any adverse conditions or non-compliance with Order No. R3-2008-0042 potentially endangering public health or the environment to the Water Board (805/549-3147), California Department of Public Health (831/655-6939), and any other agencies as appropriate within 24 hours of knowing of such conditions. A summary record of all adverse conditions or non-compliance along with corrective actions taken shall be included in each annual monitoring report.

Depending on the severity of the adverse condition or non-compliance being reported, a written report may also be required by the Water Board. The written report shall be required within five days of the initial informal reporting date and shall contain (1) a description of the non-compliance and its cause; (2) the period of non-compliance, including dates and times, and if the non-compliance has not been corrected, the anticipated time it is expected to continue; and (3) steps taken or planned to reduce, eliminate, and prevent reoccurrence of the non-compliance.

## SUPPLIER AND DISTRIBUTOR REQUIREMENTS

### I. PROVISIONS

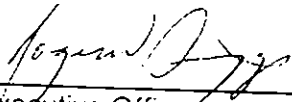
1. All quarterly monitoring shall be performed any time during the monitoring quarter (calendar quarter), but samples representative of two consecutive quarterly periods must be separated by at least one month. Monthly sampling shall be conducted at regularly scheduled times during each month and consecutive events should be approximately four weeks apart and no less than two weeks apart. Unless otherwise specified by the Monitoring and Reporting Program, annual sampling shall be performed any time during the calendar year, but samples representative of two consecutive annual periods must be obtained at least six months apart.
2. All monitoring must be conducted according to test procedures established by 40 Code of Federal Regulations Part 136, entitled, "Guidelines Establishing Test Procedures for Analysis of Pollutants." All sampling analyses shall be conducted at the lowest practical quantitation limits achievable under U.S. EPA specified methodology. In cases where effluent limits are set below the lowest achievable practical quantitation limits, constituents not detected at the practical quantitation limit will be considered in compliance with effluent limitations.
3. All samples collected shall be tracked and submitted under chain of custody and analyzed by a laboratory certified by California Department of Public Health for the specified analysis.
4. This Monitoring and Reporting Program may be revised at any time during the Permit term, as necessary, under the authority of the Executive Officer.
5. The Supplier and Distributor shall submit a copy of each monitoring report along with a completed copy of the attached monitoring and reporting program transmittal sheet to:

California Regional Water Quality Control Board  
Central Coast Region  
895 Aerovista Place, Suite 101  
San Luis Obispo, CA 93401

### J. IMPLEMENTATION

1. This monitoring and reporting program shall be implemented immediately. However the Supplier and Distributor will be allowed a two-month grace period to procure equipment and establish new sampling protocols and sampling locations as necessary for new monitoring requirements contained within this Monitoring and Reporting Program.
2. The Executive Officer may revise this MRP as appropriate.

Ordered By:

  
Executive Officer

